43. A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30°, the box starts to slip and slides 4.0 m down the plank in 4.0 s. The coefficients of static and kinetic friction between the box and the plank will be, respectively :



(1) 0.5 and 0.6

(2) 0.4 and 0.3

(3) 0.6 and 0.6

(4) 0.6 and 0.5

44. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is :

(1) $\frac{27}{5}$

(2) $\frac{5}{27}$

(3) $\frac{4}{9}$

$\frac{9}{4}$

45. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be :

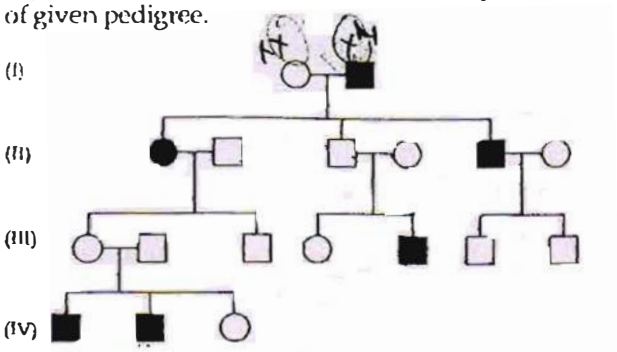
(1) $\frac{\beta^2}{\alpha}$

(2) $\frac{2\pi\beta}{\alpha}$

(3) $\frac{\beta^2}{\alpha^2}$

(4) $\frac{\alpha}{\beta}$

46. In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.



(1) Autosomal recessive

(2) X-linked dominant

(3) Autosomal dominant

(4) X-linked recessive

47. Which of the following is **not** a function of the skeletal system ?

(1) Production of body heat

(2) Locomotion

(3) Production of erythrocytes

(4) Storage of minerals

Destruction of the anterior horn cells of the spinal cord would result in loss of :

(1) commissural impulses

(2) integrating impulses

(3) sensory impulses

(4) voluntary motor impulses

49. The term "linkage" was coined by

(1) G. Mendel

(2) W. Sutton

(3) T.H. Morgan

(4) T. Boveri

50. Filiform apparatus is characteristic feature of

(1) Aleurone cell

(2) Synergids

(3) Generative cell

(4) Nucellar embryo

51. Satellite DNA is important because it :

(1) does not code for proteins and is same in all members of the population.

(2) codes for enzymes needed for DNA replication.

(3) codes for proteins needed in cell cycle.

(4) shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

52. The wheat grain has an embryo with one large, shield-shaped cotyledon known as :

(1) Scutellum

(2) Coleoptile

(3) Epiblast

(4) Coleorrhiza

53. Identify the **correct** order of organisation of genetic material from largest to smallest :

(1) Genome, chromosome, gene, nucleotide

(2) Chromosome, genome, nucleotide, gene

(3) Chromosome, gene, genome, nucleotide

(4) Genome, chromosome, nucleotide, gene

54. Most animals that live in deep oceanic waters are :

(1) tertiary consumers

(2) detritivores

(3) primary consumers

(4) secondary consumers

55. Cell wall is absent in :

(1) Mycoplasma

(2) *Nostoc*

(3) *Aspergillus*

(4) *Funaria*

56. In which of the following interactions both partners are adversely affected ?

(1) Parasitism

(2) Mutualism

(3) Competition

(4) Predation

57. Human urine is usually acidic because :

(1) potassium and sodium exchange generates acidity.

(2) hydrogen ions are actively secreted into the filtrate.

(3) the sodium transporter exchanges one hydrogen ion for each sodium ion, in peritubular capillaries.

(4) excreted plasma proteins are acidic.

58. Match the following list of microbes and their importance :

(a)	*Sacharomyces cerevisiae*	(i)	Production of immunosuppressive agents
(b)	*Monascus purpureus*	(ii)	Ripening of Swiss cheese
(c)	*Trichoderma polysporum*	(iii)	Commercial production of ethanol
(d)	*Propionibacterium sharmanii*	(iv)	Production of blood-cholesterol lowering agents

	(a)	(b)	(c)	(d)
(1)	(iv)	(ii)	(i)	(iii)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(iii)	(ii)	(i)

59. The body cells in cockroach discharge their nitrogenous waste in the haemolymph mainly in the form of :

(1) Urea

(2) Calcium carbonate

(3) Ammonia

(4) Potassium urate

60. In which of the following both pairs have **correct** combination ?

(1)	Gaseous nutrient cycle	Nitrogen and sulphur
	Sedimentary nutrient cycle	Carbon and Phosphorus
(2)	Gaseous nutrient cycle	Sulphur and Phosphorus
	Sedimentary nutrient cycle	Carbon and Nitrogen
(3)	Gaseous nutrient cycle	Carbon and Nitrogen
	Sedimentary nutrient cycle	Sulphur and Phosphorus
(4)	Gaseous nutrient cycle	Carbon and sulphur
	Sedimentary nutrient cycle	Nitrogen and Phosphorus

61. Which one is a wrong statement ?

(1) Haploid endosperm is typical feature of gymnosperms

(2) Brown algae have chlorophyll a and c, and fucoxanthin

(3) Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms

(4) *Mucor* has biflagellate zoospores

62. Match the columns and identify the correct option.

	Column I		Column II
(a)	Thylakoid		Disc-shaped sacs in Golgi apparatus
(b)	Cristae		Condensed structure of DNA
(c)	Cisterna		Flat membranous sacs in stroma
(d)	Chromatin	(iv)	Infoldings in mitochondria

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(iv)	(ii)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iii)	(iv)	(i)	(ii)

63. If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidence?
(1) Haemocytes
(2) Serum globulins
(3) Fibrinogin in plasma
(4) Serum albumins

64. The imperfect fungi which are decomposers of litter and help in mineral cycling belong to
(1) Phycomycetes
(2) Ascomycetes
(3) Deuteromycetes
(4) Basidiomycetes

65. The oxygen evolved during photosynthesis comes from water molecules. Which one of the following pairs of elements is involved in this reaction ?
(1) Magnesium and Molybdenum
(2) Magnesium and Chlorine
(3) Manganese and Chlorine
(4) Manganese and Potassium

66. Body having meshwork of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum
(1) Mollusca
(2) Protozoa
(3) Coelenterata
(4) Porifera

67. Root pressure develops due to
(1) Passive absorption
(2) Increase in transpiration
(3) Active absorption
(4) Low osmotic potential in soil

68. The enzyme that is not present in succus enteric is :
(1) nucleosidase
(2) lipase
(3) maltase
(4) nucleases

69. Metagenesis refers to :
(1) Occurrence of a drastic change in form during post-embryonic development
(2) Presence of a segmented body an parthenogenetic mode of reproduction
(3) Presence of different morphic forms
(4) Alternation of generation between asexual and sexual phases of an organism

70. A protoplast is a cell :
(1) undergoing division
(2) without cell wall
(3) without plasma membrane
(4) without nucleus

71. The DNA molecule to which the gene of interest is integrated for cloning is called
(1) Template
(2) Carrier
(3) Transformer
(4) Vector

72. Which of the following structures is not found in a prokaryotic cell ?
(1) Mesosome
(2) Plasma membrane
(3) Nuclear envelope
(4) Ribosome

73. The structures that help some bacteria to attach to rocks and/or host tissues are :
(1) Mesosomes
(2) Holdfast
(3) Rhizoids
(4) Fimbriae

74. Which one of the following hormones is not involved in sugar metabolism ?
(1) Insulin
(2) Glucagon
(3) Cortisone
(4) Aldosterone

75. In photosynthesis, the light-independent reactions take place at
(1) Photosystem II
(2) Stromal matrix
(3) Thylakoid lumen
(4) Photosystem I

76. The chitinous exoskeleton of arthropods is formed by the polymerisation of :
(1) N - acetyl glucosamine
(2) lipoglycans
(3) keratin sulphate and chond phate
(4) D -glucosamine

77. Select the wrong statement :
(1) The term *'contagium vivum fluidum'* was coined by M. W. Beijerinek
(2) Mosaic disease in tobacco and AIDS in human being are caused by viruses
(3) The viroids were discovered by D. J. Ivanowski
(4) W.M. Stanley showed that viruses could be crystallized

78. Among china rose, mustard, brinjal, potato, guava, cucumber, onion and tulip, how many plants have superior ovary ?
(1) Three
(2) Four
(3) Five
(4) Six

79. In angiosperms, microsporogenesis and megasporogenesis :
(1) Involve meiosis
(2) occur in ovule
(3) occur in anther
(4) form gametes without further div

(1) endoplasmic reticulum, ribosor
80. Cellular organelles with membranes are
(2) lysosomes, Golgi apparatus and mitochondria
(3) nuclei, ribosomes and mitochondria
(4) chromosomes, ribosomes and endoplasmic reticulum

81. Which one of the following is not applicable to RNA ?
(1) Heterocyclic nitrogenous bases
(2) Chargaff's rule
(3) Complementary base pairing
(4) 5' phosphoryl and 3' hydroxyl ends

82. Which of the following are most suitable indicators of SO_2 pollution in the environment ?
(1) Algae
(2) Fungi
(3) Lichens
(4) Conifers

83. Acid rain is caused by increase in the atmospheric concentration of :
(1) CO_2 and CO
(2) O_3 and dust
(3) SO_2 and NO_2
(4) SO_3 and CO

84. Which of the following immunoglobulins does constitute the largest percentage in human milk ?
(1) Ig A
(2) Ig G
(3) Ig D
(4) Ig M

85. Which of the following diseases is caused by a protozoan ?
(1) Babesiosis
(2) Blastomycosis
(3) Syphilis
(4) Influenza

86. Outbreeding is an important strategy of animal husbandry because it :
is useful in overcoming inbreeding depression.
(2) exposes harmful recessive genes that are eliminated by selection.
(3) helps in accumulation of superior genes.
(4) is useful in producing purelines of animals.

87. Which one of the following animals has two separate circulatory pathways ?
(1) Whale
(2) Shark
(3) Frog
(4) Lizard

88. Name the pulmonary disease in which alveolar surface area involved in gas exchange is drastically reduced due to damage in the alveolar walls.
(1) Pneumonia
(2) Asthma
(3) Pleurisy
(4) Emphysema

89. A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is
(1) Gamete internal fertilization and transfer
(2) Germ cell internal fallopian transfer
(3) Gamete inseminated fallopian transfer
(4) Gamete intra fallopian transfer

90. A gene showing codominance has :
(1) alleles that are recessive to each other
(2) both alleles independently expressed in the heterozygote
(3) one allele dominant on the other
(4) alleles tightly linked on the same chromosome

91. Doctors use stethoscope to hear the sounds produced during each cardiac cycle. The second sound is heard when :
Semilunar valves close down after the blood flows into vessels from ventricles
(2) AV node receives signal from SA node
(3) AV valves open up
(4) Ventricular walls vibrate due to gushing in of blood from atria

92. Read the different components from (a) to (d) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem :
(a) Secondary cortex
(b) Wood
(c) Secondary phloem
(d) Phellem
The correct order is :
(1) (d), (a), (c), (b)
(2) (d), (c), (a), (b)
(3) (c), (d), (b), (a)
(4) (a), (b), (d), (c)

93. Which one of the following hormones though synthesised elsewhere, is stored and released by the master gland ?
(1) Prolactin
(2) Melanocyte stimulating hormone
(3) Antidiuretic hormone
(4) Luteinizing hormone

94. The wings of a bird and the wings of an insect are :
(1) phylogenetic structures and represent divergent evolution
homologous structures and represent convergent evolution
(3) homologous structures and represent divergent evolution
(4) analogous structures and represent convergent evolution

95. Which of the following pairs is not corr matched ?

	Mode of reproduction	Example
(1)	Binary fission	*Sargassum*
(2)	Conidia	*Penicillium*
(3)	Offset	Water hyacinth
(4)	Rhizome	Banana

96. In his classic experiments on pea plants, Mendel did not use :
(1) Seed shape
(2) Flower position
(3) Seed colour
(4) Pod length

97. The function of the gap junction is to :
(1) separate two cells from each other
(2) stop substance from leaking acros a tissue.
(3) performing cementing to keep neighbouring cells together.
facilitate communication between adjoining cells by connecting the cytoplasm for rapid large molecules.
transfer of ions, small molecules and some

98. Eutrophication of water bodies leading to killing fishes is mainly due to non-availability of :
(1) essential minerals
(2) oxygen
(3) food
(4) light

99. In human females, meiosis-II is not completed until ?
(1) uterine implantation
(2) birth
(3) puberty
(4) fertilization

100. The species confined to a particular region and not found elsewhere is termed as :
(1) Endemic
(2) Rare
(3) Keystone
(4) Alien

The UN conference of Parties on climate change in the year 2012 was held at :
(1) Lima
(2) Warsaw
(3) Durban
(4) Doha

02. Which of the following layers in an antral follicle is acellular ?

(1) Stroma

(2) Zona pellucida

(3) Granulosa

(4) Theca interna

03. During biological nitrogen fixation, inactivation of nitrogenase by oxygen poisoning is prevented by :

(1) Carotene

(2) Cytochrome

(3) Leghaemoglobin

(4) Xanthophyll

04. Which of the following events is not associated with ovulation in human female ?

(1) Release of secondary oocyte

(2) LH surge

(3) Decrease in estradiol

(4) Full development of Graafian follicle

05. Arrange the following events of meiosis in correct sequence :

(a) Crossing over

(b) Synapsis

(c) Terminalisation of chiasmata

(d) Disappearance of nucleolus

(1) (a), (b), (c), (d)

(2) (b), (c), (d), (a)

(3) (b), (a), (d), (c)

(4) (b), (a), (c), (d)

06. Which of the following joints would allow no movement ?

(1) Synovial joint

(2) Ball and Socket joint

(3) Fibrous joint

(4) Cartilaginous joint

07. Flowers are unisexual in

(1) China rose

(2) Onion

(3) Pea

(4) Cucumber

108. A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind ?

(1) Nil

(2) 0.25

(3) 0.5

(4) 1

109. The cutting of DNA at specific locations became possible with the discovery of :

(1) Selectable markers

(2) Ligases

(3) Restriction enzymes

(4) Probes

110. Roots play insignificant role in absorption of water in :

(1) Pea

(2) Wheat

(3) Sunflower

(4) *Pistia*

111 Male gametophyte in angiosperms produces :

(1) Single sperm and two vegetative cells

(2) Three sperms

(3) Two sperms and a vegetative cell

(4) Single sperm and a vegetative cell

112. In which group of organisms the cell walls form two thin overlapping shells which fit together ?

(1) Dinoflagellates

(2) Slime moulds

(3) Chrysophytes

(4) Euglenoids

113. Coconut water from a tender coconut is :

(1) Innermost layers of the seed coat

(2) Degenerated nucellus

(3) Immature embryo

(4) Free nuclear endosperm

114. A column of water within xylem vessels of tall trees does not break under its weight because of :

(1) Lignification of xylem vessels

(2) Positive root pressure

(3) Dissolved sugars in water

(4) Tensile strength of water

115 Chromatophores take part in :

(1) Movement

(2) Respiration

(3) Photosynthesis

(4) Growth

116. In mammalian eye, the 'fovea' is the center of the visual field, where :

(1) only rods are present.

(2) more rods than cones are found.

(3) high density of cones occur, but has no rods.

(4) the optic nerve leaves the eye.

117. Industrial melanism is an example of :

(1) Mutation

(2) Neo Lamarckism

(3) Neo Darwinism

(4) Natural selection

118. Which of the following biomolecules does have a phosphodiester bond ?

(1) Amino acids in a polypeptide

(2) Nucleic acids in a nucleotide

(3) Fatty acids in a diglyceride

(4) Monosaccharides in a polysaccharide

119. An association of individuals of different species living in the same habitat and having functional interactions is :

(1) Ecosystem

(2) Population

(3) Ecological niche

(4) Biotic community

120. Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of :

(1) Omega 3

(2) Vitamin A

(3) Vitamin B

(4) Vitamin C

121. The introduction of t-DNA into plants involves :

(1) Exposing the plants to cold for a brief peri

(2) Allowing the plant roots to stand in water

(3) Infection of the plant by *Agrobacterium tumefaciens*

(4) Altering the pH of the soil, then heat-shocking the plants

122. Auxin can be bioassayed by :

(1) Potometer

(2) Lettuce hypocotyl elongation

(3) Avena coleoptile curvature

(4) Hydroponics

123. Pick up the **wrong** statement :

(1) Some fungi are edible

(2) Nuclear membrane is present in Monera

(3) Cell wall is absent in Animalia

(4) Protista have photosynthetic and heterotrophic modes of nutrition

124. Ectopic pregnancies are referred to as :

(1) Implantation of defective embryo in the uterus

(2) Pregnancies terminated due to hormonal imbalance.

(3) Preg nancies with genetic abnormality.

(4) Implantation of embryo at site other than uterus.

125. Axile placentation is present in

(1) Pea

(2) *Argemone*

(3) *Dianthus*

(4) Lemon

126. The primary dentition in human differs from permanent dentition in not having one of the following type of teeth

(1) Molars

(2) Incisors

(3) Canine

(4) Premolars

127. Which one of the following fruits is parthenocarpic?

(1) Jackfruit

(2) Banana

(3) Brinjal

(4) Apple

Balbiani rings are sites of :

(1) Polysaccharide synthesis

(2) RNA and protein synthesis

(3) Lipid synthesis

(4) Nucleotide synthesis

129. A pleiotropic gene :

controls a trait only in combination with another gene

controls multiple traits in an individual.

(3) is expressed only in primitive plants.

(4) is a gene evolved during Pliocene.

Grafted kidney may be rejected in a patient due to :

(1) Passive immune response

(2) Innate immune response

(3) Humoral immune response

(4) Cell-mediated immune response

131 During ecological succession :

(1) the numbers and types of animals remain constant.

(2) the changes lead to a community that is in near equilibrium with the environment and is called pioneer community

the gradual and predictable change in species composition occurs in a given area

(4) the establishment of a new biotic community is very fast in its primary phase.

132. Which of the following are **not** membrane - bound ?

(1) Lysosomes

(2) Mesosomes

(3) Vacuoles

(4) Ribosomes

133. Increase in concentration of the toxicant at successive trophic levels is known as :

(1) Biotransformation

(2) Biogeochemical cycling

(3) Biomagnification

(4) Biodeterioration

134. A jawless fish, which lays eggs in fresh water and whose ammocoetes larvae after metamorphosis return to the ocean is :

(1) *Neomyxine*

(2) *Petromyzon*

(3) *Eptatretus*

(4) *Myxine*

135. Choose the **wrong** statement :

(1) Morels and truffles are poisonous mushrooms

(2) Yeast is unicellular and useful in fermentation

(3) *Penicillium* is multicellular and produces antibiotics

(4) *Neurospora* is used in the study of biochemical genetics

136. Strong reducing behaviour of H_3PO_2 is due to

(1) High electron gain enthalpy of phosphorus

(2) High oxidation state of phosphorus

(3) Presence of two $-OH$ groups and one $P-H$ bond

(4) Presence of one $-OH$ group and two $P-H$ bonds

137. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence :

(1) Al < Ga < In < Tl

(2) ~~Tl < In < Ga < Al~~

(3) In < Tl < Ga < Al

(4) Ga < In < Al < Tl

138. The oxidation of benzene by V_2O_5 in the presence of air produces :

(1) maleic anhydride

(2) benzoic acid

(3) benzaldehyde

(4) ~~benzoic anhydride~~

139. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be :

(1) $\frac{1}{2}K$

(2) K

(3)

(4)

140. Gadolinium belongs to 4*f* series. It's atomic number is 64. Which of the following is the correct electronic configuration of gadolinium ?

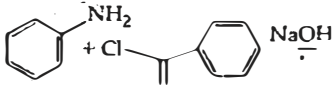
(1) [Xe] $4f^9 5s^1$

(2) [Xe] $4f^7 5d^1 6s^2$

(3) [Xe] $4f^6 5d^2 6s^2$

(4) [Xe] $4f^8 6d^2$

141. The following reaction



is known by the name

(1) Perkin's reaction

(2) Acetylation reaction

(3) Schotten-Baumen reaction

(4) Friedel-Craft's reaction

142. What is the mass of the precipitate formed when 50 mL of 16.9% so s mixed with 50 mL of 5.8% NaCl solution ?

(Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

(1) 3.5 g

(2)

(3) 14 g

(4) 28 g

143. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed ?

(1) 2.0

(2) 7.0

(3) 1.04

(4) 12.65

144. Method by which Aniline cannot be prepared

(1) degradation of benzamide with bromine in alkaline solution.

(2) reduction of nitrobenzene with H_2/Pd in ethanol.

(3 potassium salt of phthalimide treated wi chlorobenzene followed by hydrolysis wi[A] aqueous NaOH solution.

(4) hydrolysis of phenylisocyanide with acid solution

145. If Avogadro number N_A, is changed fro 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would ch

(1 the mass of one mole of carbon.

(2) the ratio of chemical species to each other in a balanced equation.

(3) the ratio of elements to each other in a compound.

(4) the definition of mass in units of grams.

146. The variation of the boiling points of the hydrogen halides is in the order HF > HI > HBr > HCl.

What explains the higher boiling point of hydrogen fluoride ?

(1) There is strong hydrogen bonding between HF molecules.

(2) The bond energy of HF molecules is greater than in other hydrogen halides.

(3) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule.

(4) The electronegativity of fluorine is much higher than for other elements in the group.

147. The number of structural isomers possible from the molecular formula C_3H_9N is :

(1) 5

(2) 2

(3) 3

(4) 4

148. Which of the statements given below is incorrect ?

(1) O_3 molecule is bent

(2) ONF is isoelectronic with O_2N^-

(3) OF_2 is an oxide of fluorine

(4) Cl_2O_7 is an anhydride of perchloric acid